ASSIGNMENT No: 3

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BATCH: CSE B3
ACADEMIC YEAR: 2025-29

TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

CODE:

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

```
class Calculator
{
    //static variable stores the number of objects created and is shared among all objects of the class
    static int count= 0;

    //constructor – name is same as class name and it is used to initialize the object of the class
    Calculator()
    {
        count++;
    }

    //Overloaded methods
    int add(int a,int b)
    {
        return a+b;
    }

    double add(double a,double b)   //double is used for decimal values, therefore a and b can used again
    {
        return a+b;
    }

    int add(int a,int b,int c)
    {
        return a+b+c;
    }

    //static method – can be called without creating an object of the class
    static void displayCount()
    {
        System.out.println("Objects created:" + count);
    }
}
public class Suhani
{
public static void main(String[] args)
{
Calculator c1 = new Calculator();
Calculator c2 = new Calculator();
System.out.println("Addition of two integers:" + c1.add(10,20));
System.out.println("Addition of two doubles:" + c1.add(12.5,20.5));
System.out.println("Addition of three integers:" + c1.add(1,2,3));
Calculator.displayCount();
}
}
```

OUTPUT:

```
Last login: Thu Jul 16 11:42:47 on ttys014
suhanigarg670@Suhanis-MacBook-Air ~ % cd desktop
suhanigarg670@Suhanis-MacBook-Air desktop % javac Suhani.java
suhanigarg670@Suhanis-MacBook-Air desktop % java Suhani
Addition of two integers:30
Addition of two doubles:33.0
Addition of three integers:6
Objects created:2
suhanigarg670@Suhanis-MacBook-Air desktop %
```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders

```java
class Restaurant
{
    static int totalOrders = 0;

    Restaurant()
    {
        totalOrders++;
    }

    double bill(double amount)
    {
        return amount;
    }

    double bill(double amount, double packingCharge)
    {
        return amount + packingCharge;
    }

    double bill(double amount, double packingCharge, double deliveryCharge)
    {
        return amount + packingCharge + deliveryCharge;
    }

    static void displayOrders()
    {
        System.out.println("Total Orders: " + totalOrders);
    }
}

public class Suhani
{
    public static void main(String[] args)
    {
        Restaurant r1 = new Restaurant();
        Restaurant r2 = new Restaurant();
        Restaurant r3 = new Restaurant();

        System.out.println("Dine-in Bill: " + r1.bill(500));
        System.out.println("Takeaway Bill: " + r2.bill(500, 20));
        System.out.println("Delivery Bill: " + r3.bill(500, 20, 50));

        Restaurant.displayOrders();
    }
}
```

OUTPUT

```
Last login: Sat Jul 18 13:21:01 on ttys027
[suhanigarg670@Suhanis-MacBook-Air ~ % cd desktop
[suhanigarg670@Suhanis-MacBook-Air desktop % javac Suhani.java
[suhanigarg670@Suhanis-MacBook-Air desktop % java Suhani
Dine-in Bill: 500.0
Takeaway Bill: 520.0
Delivery Bill: 570.0
Total Orders: 3
suhanigarg670@Suhanis-MacBook-Air desktop %
```

CONCLUSION:

This assignment successfully demonstrates the concepts of method overloading, static variables, and static methods in Java. Method overloading allows multiple methods with the same name but different parameters. Static variables and methods are shared by all objects of a class. The program also explains the concepts of classes, objects, constructors, object creation, and method calling, providing a clear understanding of object-oriented programming in Java.